PROJECT REPORT

Flash-Flood & Cloud-Burst Early-Warning Network Using LoRa

CHAPTER 1 – INTRODUCTION

1.1 Background

Flash floods and cloudbursts are major threats in hilly regions such as Himachal Pradesh. Sudden extreme rainfall causes river overflow, landslides, and destruction of life and infrastructure. Traditional systems are costly, delayed, and dependent on cellular networks, making them ineffective during emergencies. LoRa-based IoT networks provide long-range, low-power, cost-effective, and locally deployable monitoring solutions.

1.2 Problem Statement

• Flash floods occur unpredictably.

• No village-level local alert system exists.

• GSM/Wi-Fi-based systems fail during disasters.

• Need for low-cost, long-range, independent early-warning solution.

1.3 Motivation

Due to recurring disasters in Himachal Pradesh, a reliable, independent, community-level alerting network is essential. LoRa's long range (10–15 km), low power, and no-SIM requirement make it an ideal technology for such early-warning systems.

1.4 Objectives

• Develop a LoRa-based sensor network for rainfall, water level, humidity.

• Build a gateway for real-time alerting.

• Implement multi-level alerts: Safe, Caution, Warning, Emergency.

• Create a prototype and design scalability for real-world use.

CHAPTER 2 – LITERATURE REVIEW

2.1 IoT-Based Environmental Monitoring

IoT systems are widely used for rainfall, water-level, and landslide detection. GSM/Wi-Fi systems depend on infrastructure, which often fails during disasters.

2.2 LoRa Technology

LoRa provides:

• Long-range communication (up to 15 km)

• Low power consumption

• Unlicensed ISM band operation

• Ideal for remote sensor networks

Compared to Wi-Fi/GSM:

• Wi-Fi = short range, high power

• GSM = dependent on SIM, paid connectivity

• LoRa = free, independent, long-range

2.3 Flood Early-Warning Systems

Traditional systems rely on radar and satellite data, which lack locality. Literature supports community-level sensor networks for real-time alerts.

2.4 Multi-Parameter Prediction Models

Flood prediction uses rainfall, water level, humidity, and rate-of-change trends. Weighted scoring is widely adopted.

CHAPTER 3 – SYSTEM DESIGN AND WORKING

3.1 Architecture Overview

Components:

• Sensor Nodes – rainfall, water-level, humidity, temperature

• Gateway Node – receives LoRa data, computes alert, displays warnings

3.2 Hardware

• Arduino / ESP32

• LoRa SX1278 module

• Rain sensor

• Water level sensor

• DHT11 temperature & humidity sensor

• LEDs and LCD for alert output

3.3 Data Processing Pipeline

1. Sampling (30–60 seconds normal, 10–15 seconds in alert mode)

2. Data formatting with timestamp, sensor ID

3. LoRa transmission (transmitter → long-range → receiver)

4. Weighted score calculation:

Rainfall 40%

Water Level 35%

Humidity 15%

Rate-of-change 10%

5. Alert Decision:

Green, Yellow, Red, Purple

6. Multi-node intelligence (2+ RED → PURPLE)

7. Gateway alert output through LEDs/LCD

CHAPTER 4 – PROTOTYPE DEVELOPMENT & RESULTS

4.1 Prototype Implementation

• 3–4 sensor nodes placed at different heights simulating terrain

• LoRa SX1278 used for long-range transmission

• Arduino-based nodes and central gateway

• Alerts tested using LEDs and Serial Monitor

4.2 Observations

• LoRa communication stable over long distances

• Alert algorithm produced correct risk levels

• Very low power consumption suitable for solar operation

• Real-time alerting achieved (<2 seconds)

4.3 Scalability Recommendations

• Replace Arduino with ESP32 LoRa

• Add solar panels for energy

• Add cloud dashboard (ThingsBoard/Blynk)

• Integrate SMS app alerts

• Connect to district disaster management centers

CHAPTER 5 – CONCLUSION

The proposed LoRa-based flash flood and cloudburst early-warning system successfully demonstrates a low-cost, long-range, energy-efficient monitoring solution suitable for hilly regions. The prototype validates real-time alerting, reliable communication, and multi-node intelligence for flood prediction. With further enhancement using ESP32, solar power, cloud dashboards, and ML-based prediction, this system can be scaled for real-world deployment to protect vulnerable communities.

REFERENCES

1. IEEE Papers on LoRa-based disaster and environmental monitoring systems.

2. Govt. of Himachal Pradesh Flood Data (2023).

3. Semtech LoRa Technology Documentation.

4. Research articles on IoT-based flood prediction and multi-parameter sensing.

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